INCH-POUND

MIL-PRF-32643
29 January 2020

PERFORMANCE SPECIFICATION
SEALER COATING FOR MAGNESIUM ANODIZING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification governs the performance requirements to be used for qualification of coatings as sealers over anodized magnesium substrates. The sealer coating is lead and chromate free and meets the air pollution requirements for solvent emissions at 3.5 lbs/gal (420 g/l) maximum as packaged for types I and II and 1.00 lbs/gal (120 g/l) maximum as packaged for type III. The sealer coatings are free of Hazardous Air Pollutants (HAPs) and are to be used over anodizing conforming to MIL-DTL-32459. The sealer coatings are compatible with the epoxy primers and topcoats that are used in the Chemical Agent Resistant Coating (CARC) System.

1.2 Types. This specification covers the following three types of coating.

Type I	High temperature cure/bake (Above 300°F)
Type II	Low temperature cure/bake (200-250°F)
Type III	Ambient or force dry (70-120°F)

1.3 Part or identifying number (PIN). Use the following example to create the PINs to be used for coatings acquired to this specification. When other container sizes are required, change the designator accordingly, such as 1/2P for 1/2 pint.

M32643 -	X -	XXXX
Specification Identifier	Coating type designator Type I = 1 Type II = 2 Type III = 3	Container size designator 1 pint = 001P 1 quart = 001Q 1 gallon = 001G

Comments, suggestions, or questions on this document should be addressed to: Director, CCDC Army Research Laboratory, Weapons and Materials Research Directorate, Specifications & Standards Office, ATTN: FCDD-RLW-MC, Aberdeen Proving Ground, MD 21005-5069. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

AMSC N/A FSC 6850

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1.3 Safety-Hazardous Materials. Hazardous materials should be handled in accordance with all Federal, State, and local regulations.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.1.1 Specifications, standards, and handbooks. The issues of the following documents, in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

FEDERAL STANDARD

FED-STD-313	Material Safety Data, Transportation Data, And Disposal Data For Hazardous Materials Furnished To Government Activities

(Single copies of this specification, and other federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from the General Services Administration, Federal Supply Service, https://www.gsa.gov/).

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-22750	Coating, Epoxy, High Solids
MIL-PRF-23377	Primer Coatings: Epoxy, High-Solids
MIL-DTL-32459	Coatings, Anodic for Magnesium and Magnesium Alloys
MIL-PRF-23699	Lubricating Oil, Aircraft Turbine Engine, Synthetic Base
MIL-DTL-53039	Coating, Aliphatic Polyurethane, Single-Component, Chemical Agent Resistant
MIL-DTL-64159	Camouflage Coating, Water Dispersible Aliphatic Polyurethane, Chemical Agent Resistant
MIL-PRF-83282	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Metric
MIL-PRF-83936	Remover, Paint, Tank Type, Landing Gear Components, and Other Aircraft and Support Equipment
MIL-PRF-85582	Primer Coatings: Epoxy, Waterborne
DOD-PRF-85734	Lubricating Oil, Helicopter Transmission System, Synthetic Base

(Copies of these documents are available online at https://quicksearch.dla.mil/).

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2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

UNITED STATES ENVIRONMENTAL PROTECTION AGENCY (EPA)

EPA Method 311	Analysis of Hazardous Air Pollutant Compounds in Paints and Coatings by Direct Injection into a Gas Chromatograph

(Copies of this document are available online at https://www.epa.gov/emc/method-311-hazardous-air-pollutant-compounds-paints-and-coatings).

2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on the date of invitation for bids or request for proposal should apply.

ASTM INTERNATIONAL

ASTM B117	Standard Practice for Operating Salt Spray (Fog) Apparatus.
ASTM D2247	Standard Practice for Testing Water Resistance of Coatings in 100% Relative Humidity
ASTM D2794	Standard Test Method for Resistance of Organic Coatings to the Effects of Rapid Deformation (Impact)
ASTM D3359	Standard Test Method for Measuring Adhesion by Tape Test.
ASTM D3363	Standard Test Method for Measuring Film Hardness by Pencil Test
ASTM D3924	Standard Specification for Standard Environment for Conditioning and Testing Paint, Varnish, Lacquer, and Related Materials
ASTM D3960	Standard Practice for Determining Volatile Organic Compound (VOC) Content of Paints and Related Coatings
ASTM D6886	Standard Test Method for Determination of the Weight Percent Individual Volatile Organic Compounds in Waterborne Air-Dry Coatings by Gas Chromatography
ASTM D4541	Standard Test Method for Pull-Off Strength of Coatings Using Portable Adhesion Testers

(Copies of these documents are available from https://astm.org/.)

OSHA Regulations - Hazardous Communication Standards - 29 CFR 1910.1200

(Copies of these documents are available from https://www.OSHA.gov.)

SOCIETY OF AUTOMOTIVE ENGINEERS INTERNATIONAL

SAE AMS-STD-595	Colors Used in Government Procurement

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(Copies of these documents are available from https://www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The coating furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable Qualified Products Database (QPD) at the time set for opening of bids (see 4.2 and 6.4). Any change in the formulation or processing of a qualified product shall necessitate its requalification. The material supplied under the contract shall be identical, within manufacturing tolerance, to the product that received qualification.

3.2 Materials. The sealer coatings furnished under this specification, when mixed and applied in accordance with the manufacturer's instructions, shall produce a coating that satisfies all of the requirements of this specification. The composition of the coatings or compounds furnished under this specification shall be the responsibility of the manufacturer, except as limited by this specification. The type I and II sealer coatings shall be an organic stoving enamel. The type III sealer coating shall be an inorganic water base sol-gel or silane. The color of the final seal coat shall be translucent in a color that differentiates this application from the magnesium anodizing conforming to MIL-DTL-32459.

3.3 Composition (Solvent analysis for VOC determination). When mixed per manufacturer's instructions, the product shall meet the applicable requirements of this specification. The seal coat shall be hazardous air pollutants-free (HAPs-free) and have a maximum Volatile Organic Content (VOC) content of 3.5 lbs/gal (420 g/l) as packaged for types I and II and a maximum VOC content of 1.00 lbs/gal (120 g/l) as packaged for type III (see 4.6.2).

3.3.1 Pigment. Hexavalent chromium shall not be present in any type. Pigments contained in these primers shall be free of HAPs and regulated toxic metals. Pigments shall not contain compounds of antimony, arsenic, beryllium, cadmium, cobalt, chromium, cyanide, lead, manganese, mercury, nickel, and selenium. Silica flattening pigments and tinting pigments are permissible for type III.

3.3.2 Volatile organic hazardous air pollutant-free (VOHAP-free). The seal coat shall be VOHAP-free when tested in accordance with 4.6.2.1.

3.4 Qualification requirements.

3.4.1 Condition in container. When tested as specified in 4.6.3, a freshly opened full container of the coating shall be free from grit, coarse particles, skins, lumps, seeds, livering or abnormal thickening. The coating shall show no pigment settling or caking that cannot be readily reincorporated to a smooth homogeneous state.

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3.4.2 Storage stability.

3.4.2.1 Accelerated storage stability. After testing as in 4.6.4.1, the coating shall meet all the requirements of the specification. There shall be no curdling or hard dry caking and any sedimentation shall easily mix back into a smooth homogeneous state.

3.4.2.2 Full container storage stability. The liquid coating, stored in the full quart can or submitted touch up kit, shall show no skinning, livering, curdling, hard dry caking nor tough gummy sediment when tested as specified in 4.6.4.2. It shall remix readily to a smooth homogeneous state and shall meet all other requirements of this specification.

3.4.3 Spraying properties. When tested as specified in 4.6.5, the coating shall show no running, sagging or streaking. The cured film shall show no dusting, mottling, color separation, flooding or floating, and shall present a smooth, uniform finish free from defects. The appearance of tears, pinholes, blisters, and bubbles are considered non-acceptable and will constitute failure.

3.4.4 Brushing properties. When tested as specified in 4.6.6, the coating shall brush satisfactorily and shall dry to a smooth, uniform film, free from seeds, runs, sags, or streaks. The dried film shall show no discernible brush marks.

3.4.5 Cure . When tested as specified in 4.6.7, there shall be no evidence of paint removal down to the anodized coated surface. This is evidence of an uncured film.

3.4.6 Coating removal. When tested as specified in 4.6.8, coating shall be attacked and lifted from 95% of the surface for two out of three panels.

3.4.7 Adhesion.

3.4.7.1 Dry tape adhesion. The assessment of the adhesion of the coating film shall be determined by its ability to not peel from the substrate when tested in accordance with ASTM D3359 as specified in 4.6.9.1. The resultant test rating shall be classified as scale 4B or better.

3.4.7.2 Wet tape adhesion. There shall be no peeling or delamination between the primer and topcoat or at the primer and substrate interface. The coatings shall have a rating of no less than a 4A when examined in accordance with ASTM D3359 (see 4.6.9.2).

3.4.7.3 Pull-off adhesion. When tested as specified in 4.6.9.3, the assessment of the pull off adhesion shall have a minimum of 1500 pounds per square inch (psi) pull strength.

3.4.8 Recoatability. When tested as specified in 4.6.10, recoating of a dried film shall produce no lifting, softening, or other film irregularity.

3.4.9 Immersion fluid resistance. When tested as specified in 4.6.11, a film of the coating shall show no blistering or wrinkling when examined immediately after removal from the test fluid. When examined 2 hours after removal, there shall be no excessive softening, whitening, or dulling. Film softening shall not exceed a 2 pencil hardness difference (see ASTM D3363) from an unexposed film with identical cure history prior to hydrocarbon exposure. After 24 hours drying, the portion of the panel which was immersed shall be almost indistinguishable with

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regard to hardness, adhesion, and general appearance from a panel prepared at the same time but not immersed. The panels, after 24 hour drying, shall comply with the adhesion requirements in 3.4.7.1.

3.4.10 Salt spray resistance. Panels to be tested shall be coated with primer MIL-PRF-23377, Class N and CARC topcoat MIL-DTL-53039 or MIL-DTL-64159. When tested as specified in 4.6.12 and examined immediately after removal from the salt spray test, films of the coating shall have no surface corrosion.

3.4.11 Humidity resistance. When tested as specified in 4.6.13, and examined immediately after removal from the humidity test, panels shall have no surface corrosion or blistering in the field away from the scribe. In addition, there shall be no corrosion of the underlying metal for a distance of more than ¼ inch from the scribe. After 4 hours air drying, the panel shall be almost indistinguishable with regard to adhesion and hardness from an unexposed film with identical cure history prior to exposure.

3.4.12 Impact Resistance. The primer coatings shall exhibit an elongation of not less than 10 percent when tested in accordance with 4.6.14. There shall be no cracking, chipping or loss of adhesion of the coating system from the magnesium-anodized panels.

3.4.13 Toxic ingredients. The seal coating shall contain no benzene (benzol), chlorinated solvents, or ethylene based glycol ethers and their acetates (see 4.6.15). The coating shall have no adverse effects on the health of personnel when used for its intended purpose.

3.5 User instruction marking and precaution sheet. All primary containers shall be legibly marked or labeled with the VOC content and VOHAP content in lbs/gal or g/l of coating. For multiple component kits, all primary containers shall be marked or labeled with the component (A, B or C) and the content (resin component, catalyst component or solvent), as applicable with the mixing and thinning instructions. In addition, a printed precaution sheet will be supplied with the following information:

CAUTION: The Surgeon General requires airline respirators to be used unless air sampling shows exposure to be below standards. Then, either chemical cartridge respirators or airline respirators are required. Avoid contact with skin and eyes. Use adequate ventilation. For other safety recommendations, refer to the Safety Data Sheet (SDS). Keep containers closed.

3.6 SDS. The manufacturer shall comply with the requirements as specified in the OSHA Brief (see: https://www.osha.gov/dsg/hazcom/) and as specified in Appendix D of 29 CFR 1910.1200). (see: http://www.osha.gov/pls/oshaweb/owadisp.show_document?p_table=standards&p_id=10103). A SDS shall be prepared for the material in accordance with FED-STD-313 and forwarded to the qualifying activity (see 6.4). The SDS shall be included with each shipment of the material covered by this specification and submitted to pertinent Government agencies as stated in FED-STD-313.

3.7 Toxicity clearance. All new chemicals and materials being added to the Army supply system shall have a toxicity clearance. A toxicity clearance involves a toxicological evaluation of materials prior to introduction into the Army supply system. The Army program manager shall

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be responsible for identifying technically feasible materials and requesting a toxicity clearance for use of that material within their program (see 6.6).

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2 and 6.4).
b. Conformance inspection (see 4.3).

4.2 Qualification inspections. Qualification inspections shall be conducted by the qualifying activity (see 6.4). The qualification test sample shall be as specified by the qualifying activity. The samples shall be legibly identified (see 6.4.4). Qualification inspections shall consist of tests for all requirements specified in section 3 and table I. Qualification inspections shall be examined for user instruction markings (see 3.5). The results of each test shall be compared with the applicable requirement in section 3. Failure to conform to any requirement shall be counted as a defect and the coating represented by the sample test shall not be approved for inclusion in the Qualified Products Database (QPD) under this specification.

4.3 Conformance inspection. The manufacturer shall forward from each production lot (see 4.3.1) a batch validation letter detailing the batch number, manufacturer's code, specification and type number, QPD number, and batch volume to CCDC Army Research Laboratory, ATTN: FCDD-RLW-MC, Camouflage, Coatings and Corrosion Team, 6300 Rodman Road, Building 4600, Aberdeen Proving Ground (APG), MD 21005-5069. The manufacturer shall perform conformance inspection testing on each production lot and have these test results on file and provided when requested by the contracting officer (see 6.2). Conformance inspections shall consist of tests for condition in container, spraying properties, VOC content and HAP-content. There shall be no failures (see 6.5).

4.3.1 Lot and batch formation. For purposes of conformance inspection, a lot shall consist of all coatings of the same type and composition from a single uniform batch, produced and offered for delivery at one time. A batch shall consist of all coating material (in U.S. gallons) manufactured during one continuous operation and forming part of one contract or order for delivery. When required, the manufacturer shall furnish with each lot and/or batch a certified test report showing that the material has passed the conformance inspection, and that there has been no formulation or process change from that which resulted in the production of the qualification inspection sample. The addition of any substance that was not present in the qualification sample requires requalification.

4.4 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with ASTM D3924. Also, unless otherwise stated in the test method or paragraph, room temperature shall be 73 ± 4°F (23 ± 2°C) and a 40 - 70 percent relative humidity.

4.5 Safety Data Sheet. The SDS shall address all individual components as they are supplied and be in compliance with 3.6. Noncompliance to 3.6 shall be cause for rejection.

4.6 Test methods.

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4.6.1 Test conditions. Except as otherwise specified herein, the routine testing and referee testing shall be conducted in accordance with ASTM D3924. A dry film thickness of 1.5± 0.2 mils (37.5 ± 5 microns) shall be used whenever film thickness is requested in any test, unless otherwise required by the test or specified by the manufacturer. The total dry film thickness of the sealer system shall not exceed the 1.5 ± 0.2 mils (37.5 ± 5 microns). Failure of any test result to fall within the specified ranges in section 3 shall constitute failure of the applicable test.

4.6.1.1 Test procedures. Tests (see table I) shall be conducted in accordance with ASTM, EPA, Society of Automotive Engineers (SAE) International or as specified herein. The right is reserved to make any additional tests deemed necessary to determine that the coating meets the requirements of this specification.

4.6.1.2 Test panels. Unless otherwise specified, test panels utilized herein shall be made from the magnesium alloy, ZE41A, with an anodic coating applied per MIL-DTL-32459. For impact testing, the panel thickness shall be a minimum of 3/8 inch (0.95 cm) thick. The type III coating shall have a clear or translucent layer of MIL-PRF-22750 producing a maximum dry film thickness of 1.5 mils (37.5 microns)

4.6.1.3 Epoxy primer. When an epoxy primer is required by the test method, apply epoxy primer conforming to MIL-PRF-23377, type I, class N and let air dry for fourteen (14) days. Apply the epoxy primer to a total dry film thickness of 0.60 to 0.90 mils (15 to 23 microns). For recoatability and adhesion testing (dry tape, wet tape, and pull-off), panels will be prepared with both MIL-PRF-23377, as specified above, and MIL-PRF-85582, type I, class N. The MIL-PRF-85582 will also be applied at a total dry film thickness of 0.60 to 0.90 mils (15 to 23 microns) and let air dry for fourteen (14) days..

4.6.1.4 Polyurethane topcoat. When a topcoat is required by the test method use a polyurethane coating conforming to either MIL-DTL-53039 or MIL-DTL-64159, Tan 686A, color number 33446 of SAE AMS-STD-595, Colors Used in Government Procurement. Apply the coating to a total dry film thickness of 2 ± 0.2 mils (50 ± 5 microns). If applied in two coats, allow the first coat to air dry for 60 minutes prior to application of the second coat. After application of the topcoat to the required thickness and prior to testing, allow the coating to air dry for not less than 168 hours.

4.6.2 Composition (Solvent analysis for VOC determination). Determine the VOC content in accordance with ASTM D3960. Check for compliance with 3.3.

4.6.2.1 VOHAP content of coatings. Hazardous solvent content of each individual coating shall be determined in accordance with ASTM D6886 or EPA Method 311, as applicable. Check for compliance with 3.3.2.

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TABLE I. Test methods.

Item	ASTM Test Method	Test Paragraph	Requirement Paragraph
Composition	D3960	4.6.2	3.3
Pigment	-	-	3.3.1
VOHAP 1/	D6886	4.6.2.1	3.3.2
Condition in container	-	4.6.3	3.4.1
Accelerated storage stability	-	4.6.4.1	3.4.2.1
Full container storage stability	-	4.6.4.2	3.4.2.2
Spraying properties	-	4.6.5	3.4.3
Brushing properties	-	4.6.6	3.4.4
Cure	-	4.6.7	3.4.5
Coating removal	-	4.6.8	3.4.6
Dry tape adhesion	D3359	4.6.9.1	3.4.7.1
Wet tape adhesion	D3359	4.6.9.2	3.4.7.2
Pull-off adhesion	D4541	4.6.9.3	3.4.7.3
Recoatability	-	4.6.10	3.4.8
Immersion fluid resistance	D3363	4.6.11	3.4.9
Salt spray resistance	B117	4.6.12	3.4.10
Humidity resistance	D2247	4.6.13	3.4.11
Impact Resistance	D2794	4.6.14	3.4.12

1/ EPA Method 311.

4.6.3 Condition in container. Upon initially opening of a full previously unopened container, the condition of the contents shall be examined for compliance with 3.4.1.

4.6.4 Storage stability.

4.6.4.1 Accelerated storage stability. Fill an epoxy lined pint container with the coating. Tightly seal the container and place sample in a preheated oven for 168 hours at 140 ± 1.8°F (60 ± 1°C). Allow to cool to room temperature and examine the contents. During the storage period, it is advised that the unopened containers be placed in larger, vented containers to confine any splash that may occur if the lid of the unopened container is blown off by gassing. If, upon removal, the unopened container is deformed, do not open. If the container is not deformed, open carefully and examine its contents. Check for compliance with 3.4.2.1.

4.6.4.2 Full container storage stability. Allow a full 8 ounce can of the coating to stand for one (1) year under laboratory room conditions of 73 ± 4°F (23 ± 2°C) and a 40 - 70 percent relative humidity. At the end of this period, examine the contents for compliance with 3.4.2.2.

4.6.5 Spraying properties. Apply coating according to manufacturer's recommendations to the anodized magnesium panels described in 4.6.1.2. If necessary for application, follow the

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manufacturer's instructions for thinning not to exceed the VOC level (see 1.1). The type III shall have a layer of MIL-PRF-22750 applied, as specified in 4.6.1.2. Cure according to manufacturer's instructions. Observe the spraying properties for compliance with 3.4.3.

4.6.6 Brushing properties. Apply the coating according to the manufacturer's recommendations to panels described in 4.6.1.2 using a 2-1/2 inch brush. Check for compliance with 3.4.4.

4.6.7 Cure. Prepare a panel as specified in 4.6.5. Thoroughly wet a lint free rag with acetone or methyl ethyl ketone (MEK) and briskly rub the painted surface for ten seconds to remove any dry spray or overspray. Wet another clean dry rag with acetone or MEK and briskly rub the same area with 50 strokes approximately six inches (15.24 cm) in length. Examine for compliance with 3.4.5.

4.6.8 Coating removal. Prepare paint strippers per manufacturer's recommendations using the temperature ranges given in table 1 of MIL-PRF-83936. Take three (3) panels prepared as in 4.6.9.1 with primer and topcoat and immerse in a container of the paint remover. The maximum dwell time is 24 hours for each paint stripper. The operating temperature is per manufacturer's instructions. At the conclusion of the test period, the panels shall be rinsed with a water spray (hot or cold) not exceeding 100 psi. The test panels shall be air dried and examined for compliance with the requirements of 3.4.6. See 6.6 for Fluorescent Penetrant Inspection requirements.

4.6.9 Adhesion.

4.6.9.1 Dry tape adhesion. Prepare nine (9) panels as specified in 4.6.5. Apply primer per 4.6.1.3 to all of these panels and air dry for fourteen (14) days. Using three (3) of these panels, perform adhesion testing as specified in ASTM D3359, method B. Examine for compliance with 3.4.7.1.

4.6.9.2 Wet adhesion. Using three (3) panels prepared in 4.6.9.1, they shall be immersed in distilled water for 24 hours at room temperature (see 4.4). Remove the test panels from the water and wipe dry with a soft cloth. Within 3 minutes after removal from the water make two parallel scribes with a stylus through the coating to the substrate. The scribes shall be ¾ of an inch apart and 2 inches long. The panels shall then be scribed to the substrate from opposing ends of the parallel scribes to form an "X". Immediately apply a 1-inch wide strip of masking tape with the adhesive side down across the scribes. Press the tape against the surface of the coating by passing a 4-1/2-pound rubber covered roller, approximately 3-1/2 inches in diameter and 1-3/4 inches in width across the tape eight times. The masking tape shall have an average adhesion of at least 60- ounces/inch width (3M Company #250 or equivalent as approved by the qualifying activity). The tape shelf life is typically one year from date of manufacture. Perform adhesion testing as specified in ASTM D3359, method B and examine for compliance with 3.4.7.2.

4.6.9.3 Pull-off adhesion. Using the remaining three (3) panels prepared in 4.6.9.1, a minimum of ten (10) pull-off tests shall be completed on each panel (top and bottom). Testing according to ASTM D4541 shall be completed using a hydraulic pull-off adhesion tester with a minimum upper limit of 3,500 psi. Testing shall be done to failure or upper limit of device. Inspection shall note the final pressure and the failure mode. Examine for compliance with 3.4.7.3.

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4.6.10 Recoatability. To a panel coated and prepared as in 4.6.5, apply primer per 4.6.1.3 and air dry for fourteen (14) days. Examine for lifting, softening, and evidence of other film irregularity for compliance with 3.4.8.

4.6.11 Immersion fluid resistance. Prepare twelve (12) panels as specified in 4.6.5. Do not wax or coat the exposed metal surfaces. Immerse three (3) of these panels under each condition as specified below. Panels shall be immersed at a minimum depth of 50 %. At the end of the test period, remove and examine for compliance with 3.4.9. Nonconformance to 3.4.9 shall constitute failure of this test.

a) F24 or JP8 at 70 ± 5°F (21 ± 3°C) for 48 hours
b) Lubricating oil conforming to either MIL-PRF-85734 or MIL-PRF-23699 at 250± 5°F (121± 5°C) for 30 days
c) Lubricating oil conforming to either MIL-PRF-85734 or MIL-PRF-23699 at 300± 5°F (149± 5°C) for 48 hours
d) Hydraulic fluid conforming to MIL-PRF-83282 at 150± 5°F (65± 5°C) for 48 hours.

4.6.12 Salt spray resistance. Prepare five (5) test panels as per 4.6.9.1. Allow to air dry for fourteen (14) days. Ensure that all panels have their edges well protected with wax or other suitable coating, but do not score. The panels shall be exposed to a 5 percent salt spray fog test in accordance with ASTM B117 for 3000 hours. Upon completion of the test, panels shall be removed and washed gently in running water, no warmer than 100°F (38°C), until surface is free from any visible salt deposits and then immediately examined for compliance with 3.4.10. Nonconformance with 3.4.10 shall constitute failure of this test.

4.6.13 Humidity resistance. Ensure that five panels prepared as in 4.6.9.1 have their edges well protected with wax or other suitable coating. Each specimen shall be scribed with a single vertical line down the center of the panel. The scribe shall be a minimum of ½ inch away from the edge of the panel. Expose panels to 3000 hours of humidity in accordance with ASTM D2247. Panels shall be examined for compliance to 3.4.11.

4.6.14 Impact Resistance. To a panel coated and prepared as in 4.6.5, apply primer per 4.6.1.3 and air dry for fourteen (14) days. The coating system shall be tested at the room temperature and in accordance with ASTM D2794 using a Gardco GE Universal Impact Tester, Model #172 (or equivalent as approved by the qualifying activity) (see 6.7). Place the coated panel, film side downward, on the rubber pad at the bottom of the impactor guide. Using a two lbs (907 gram) weight, drop the impactor 40 inches (102 cm) on the panel through the impactor guide, ensuring that the impression of the entire rim of the impactor is made in the panel. Reverse the impactor ends and drop it through the guide on the panel adjacent to the first area of impact. Using 10 power magnification, examine for conformance to 3.4.12; record the percent elongation corresponding to the largest spherical impression at which no cracking occurs. Panels shall be examined for compliance with 3.4.12

4.6.15 Toxic ingredients. The manufacturer shall certify that the coating contains no benzene, chlorinated solvents or ethylene based glycol ethers and their acetates. Check for compliance with 3.4.13. Nonconformance with 3.4.13 shall constitute failure of this test.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity. For markings, containers or packaging of the coating shall be marked with a label that identifies the specification number, type, color, and material designation so that traceability to the QPD can be verified.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The sealer coatings described in this specification, which are lead and chromate free, are intended for use on Plasma Electrolytic Oxidation (PEO) anodized magnesium surfaces in CARC coating system. The seal coatings, depending upon a bake or air-dry applicatio10n, will meet a 3.5 lbs/gal (420 g/l) or a 1.00 lbs/gal (120 g/l) maximum VOC content requirement and be HAP-free. The coatings qualified to this specification encompass high bake, low bake and air dry/force dry application. The anodizing to be sealed by the coatings described conform to military specification MIL-DTL-32459.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.
b. Type of coating required (see 1.2).
c. Part or identifying number (PIN) (see 1.3).
d. Required markings (see 3.5).
e. Size of primary container.
f. Whether a SDS is required with each shipment (see 3.6 and 4.5).
g. If a toxicity clearance is required (see 3.7).
h. If qualification samples are required and where to send them (see 4.2 and 6.4).
i. If conformance samples are required and where to send them (see 4.3).
j. Lot and batch formation (see 4.3.1).
k. Packaging requirements (see 5.1).

6.3 Basis of purchase. The coating covered by this specification should be purchased by volume, the unit being one U.S. liquid gallon of 231 cubic inches at 68°F (20°C).

6.4 Qualification inspection. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in the QPD, whether or not such products have actually been so listed by that date. The attention of

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contractors is called to this requirement and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Samples for QPD testing (see 4.2) and for the validation testing program (see 4.3) should be submitted to CCDC Army Research Laboratory, ATTN: FCDD-RLW-MC, Camouflage, Coatings and Corrosion Team, 6300 Rodman Road, Building 4600, APG, MD 21005-5069. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil/.

6.4.1 Retention of qualification. In order to retain qualification of a product approved for listing in the QPD, the manufacturer will verify by certification to the qualifying activity that the manufacturer's product complies with the requirements of this specification. Unless otherwise specified, the time of periodic verification by certification will be in two year intervals from the date of the original qualification, and will be initiated by the qualifying activity. No change will be made in formulation, raw materials or supplier(s) of raw materials, methods of manufacture, equipment, or geographic location without prior written Government approval. The Government reserves the right to re-examine the qualified product whenever deemed necessary to determine that the product continues to meet any or all of the specification requirements.

6.4.2 Conformity to qualified sample. All lots of coatings supplied under this specification must be manufactured using the same formulation, raw materials and supplier(s) of raw materials, methods of manufacture, equipment, and geographic location as the qualification sample, unless changes have been approved by the qualifying activity.

6.4.3 SDS. Contracting officers will identify those activities requiring copies of a completed SDS prepared in accordance with 3.6. The pertinent Government mailing addresses for submission of data are listed in 3.6.

6.4.4 Sample identification for qualification inspection. Samples for QPD testing are to be identified in a cover letter with the following information:

a. Manufacturer's name and product number. Submitted by (name and date).
b. Specify the number of samples.
c. Specify the reason for submitting the samples.
d. Specification MIL-PRF-32643; Type___
e. 'SEALER COATING FOR MAGNESIUM ANODIZING"
f. Provide a copy of the SDS.
g. Provide a copy of the notarized statement of composition.
h. Provide a copy of the technical data sheet.
i. Provide a copy of the test report.

6.5 Conformance rejection and retest. Failure in any conformance inspection will result in the rejection of the batch from which it was obtained and constitutes justification for removal from the qualified products list. Rejected material cannot be resubmitted for acceptance without written approval from the qualification activity (see 4.3). The application for resubmission will contain all details concerning previous rejections and measures taken to correct these deficiencies.

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6.6 Fluorescent Penetrant Inspection. Stoving enamels and coatings approved to this specification should be removed from the substrate prior to the Fluorescent Penetrant Inspection.

6.7 Toxicity clearance request. Department of the Army Regulation (AR) 40-5, Preventive Medicine, (AR) 70-1, Acquisition Policy, and Department of the Army Pamphlet 70-3, Acquisition Procedures, require a toxicity clearance. Army toxicity questions and/or a toxicity clearance request should be addressed to: Commander, US Army Center for Health Promotion and Preventive Medicine, ATTN: MCHB-TS-TTE, 5158 Blackhawk Road, APG, MD 21010-5403.

6.8 Impact tester source. An instrument for conducting the impact resistance (see 4.6.14) is the Gardco GE Universal Impact Tester, Model #172, available from the Paul N. Gardner Company, 316 NE First Street, PO Box 10688, Pompano Beach, FL 33061-6688. There may be other instruments equivalent to this unit.

6.8 Subject term (key word) listing.

Anodized
CARC
HAP
VOC
VOHAP

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CONCLUDING MATERIAL

Custodians:
Army - MR
Navy - SH
Air Force - 20
DLA - GS

Review activities:
Army – CR, CR4, MD1, MI
Navy - AS, CG, MC
Air Force - 68

Civil agency:
GSA/FAS

Preparing activity:
Army – MR

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NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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